



Service Bulletin Number: 4021403

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Signature/ISX HPI Rev 1 to Rev 2 Fuel System Conversion

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Introduction

This publication contains information regarding the correct procedure to convert a 1998/99 Australia (ADR) Certified ISX Family 1 (ISX engines with horsepower rating between 400 and 465 hp) or 1998/99 ISX Family 3/Signature engine (ISX engines with horsepower ratings above 525 hp) from the Revision 1 HPI fuel system to the Revision 2 HPI fuel system. This procedure was developed in order to address the issue of parts availability of the Revision 1 fueling or metering actuators and injectors. In the event that a Revision 1 fueling or timing actuator, or injector is **not** available for service needs, this procedure **must** be followed in order to repair the engine correctly and maintain emissions standards compliance. This procedure will comply with Field Fix Number 316 listed in the Control Parts List, Bulletin 4021326.

Note : Wiring kit, Part Number 3164855, must be used during this repair.

Below is a brief summary of the component changes required. Refer to the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239, for additional procedures required to convert Australia (ADR) ratings.

Note : Australia (ADR) conversions must only be performed during extensive engine repairs or engine rebuild, due to the steps necessary to perform the conversion in these ratings.

Conversion Summary

- 1998 to 1999 engine position sensor (EPS)
 - EPS (see Table 1A)
 - Injector camshaft (see Table 1A)
 - Crankshaft (see Table 1A)
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1A)
 - Metering and timing actuators (see Table 1A)
- ADR 450/1650 calibration for CPL 2814 (see Table 2A)
- Modify dataplate to support 2814 CPL and field fix number 316.
- 1998 to 1999 engine position sensor (EPS)

- EPS (see Table 1B)
- Injector camshaft (see Table 1B)
- Crankshaft (see Table 1B)
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1B)
 - Metering and timing actuators (see Table 1B)
- Turbocharger (see Table 1B)
- ADR 450/1650 calibration for CPL 8048 (see Table 2B)
- Modify dataplate to support 8048 CPL and field fix number 316
- Injector camshaft (see Table 1C)
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1C)
 - Metering and timing actuators (see Table 1C)
- ADR 450/1650 calibration for CPL 2814 (see Table 2C)
- Modify dataplate to support 2814 CPL and field fix number 316
- Injector camshaft (see Table 1D)
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1D)
 - Metering and timing actuators (see Table 1D)
- Turbocharger (see Table 1D)
- ADR 450/1650 calibration for CPL 8048 (see Table 2D)
- Modify dataplate to support 8048 CPL and field fix number 316
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1E)
 - Metering and timing actuators (see Table 1E)
- Calibration for correct CPL (see Table 3)
- Modify dataplate to support new CPL and field fix number 316
- Revision 1 to Revision 2 fuel system
 - Fuel injectors (see Table 1F)
 - Metering and timing actuators (see Table 1F)
- Calibration for correct CPL (see Table 4)
- Modify dataplate to support new CPL and field fix number 316

**Table 1A - CPL
2440 to CPL
2814 Required
Parts List**

Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Sensor, Position	3331017	4001902	4001902	2
Camshaft, Injector	3680778	3681710	3681710	1

Table 1A - CPL 2440 to CPL 2814 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Crankshaft	3681073	3608889 or 3681911	3608889 or 3681911	1
Injector	4001974 or 4009674 or 4010292	4010225 or 4062567	4010225NX or 4010225RX or 4062567NX or 4062567RX	6
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Kit, Wiring	N/A	N/A	3164855	1
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				
Table 1B - CPL 2440 to CPL 8048 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Sensor, Position	3331017	4001902	4001902	2
Camshaft, Injector	3680778	3681710	3681710	1
Crankshaft	3681073	3608889 or 3681911	3608889 or 3681911	1
Injector	4001974 or 4009674 or 4010292	4010225 or 4062567	4010225NX or 4010225RX or 4062567NX or 4062567RX	6

Table 1B - CPL 2440 to CPL 8048 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Turbocharger	3591788	3595090	4025184NX or 4025184RX	1
Kit, Wiring	N/A	N/A	3164855	1
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				
Table 1C - CPL 2803 to CPL 2814 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Camshaft, Injector	3680778	3681710	3681710	1
Injector	4001974 or 4009674 or 4010292	4010225 or 4062567	4010225NX or 4010225RX or 4062567NX or 4062567RX	6
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Kit, Wiring	N/A	N/A	3164855	1

Table 1C - CPL 2803 to CPL 2814 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				
Table 1D - CPL 2803 to CPL 8048 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Camshaft, Injector	3680778	3681710	3681710	1
Injector	4001974 or 4009674 or 4010292	4010225 or 4062567	4010225NX or 4010225RX or 4062567NX or 4062567RX	6
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Turbocharger	3591788	3595090	4025184NX or 4025184RX	1
Kit, Wiring	N/A	N/A	3164855	1

Table 1D - CPL 2803 to CPL 8048 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				
Table 1E - CPL 2442 to CPL 8172 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Injector	4001791 or 4009672 or 4010291 or 3348422	4062568	4062568NX or 4062568RX	6
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Kit, Wiring	N/A	N/A	3164855	1
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				

Table 1F - CPL 2630/2801 to CPL 2834/2836 Required Parts List				
Description	Original Part Number	New Part Number	Service Part Number	Quantity Required
Injector	4001791 or 4009672 or 4010291 or 3348422	4062568	4062568NX or 4062568RX	6
Actuator, Timing	3348771	4902738	4089540RX	2
Actuator, Metering	4001835	4088780	4089428RX	2
Kit, Wiring	N/A	N/A	3164855	1
NOTE: These parts are required parts for the conversion. Other parts can possibly be necessary during the repair, such as gaskets, etc.				

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L 2 8 1 4 C a l i b r a t i o n M a t r i x

S C	D O	F R	A P	M o d e l N a m e	A d v. (h p)	A d v. (r p m)	T · P. (l b - f t)	T · P. (r p m)	R a t e d (h p)	R a t e d (r p m)	G o v. (h p)	G o v. (r p m)	C P L (P P)	P W	F C	E C M C o d e f o r 3 6 8 0 5 0 9	E C M C o d e f o r 3 4 0 8 4 2 6	E C M C o d e f o r 3 6 8 1 4 0 5	E C M C o d e f o r 3 4 0 8 5 0 1	C o m m e n t s
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A blank coordinate grid with a horizontal axis labeled x and a vertical axis labeled y . The grid consists of 10 columns and 10 rows of squares. The x -axis is labeled from 1 to 10, and the y -axis is labeled from 1 to 10.

1 0 3 2 9	1 0 7 8	1 0 1 9 3	1 0 2 7	I S X - 4 5 0	4 5 0	N / A	1 6 5 0	1 2 0 0	4 5 8	1 7 0 0	4 5 0	1 8 0 0	2 4 4 0	1 0 0 8	P 0 2 2	N 1 0 0 2 1	N / A	N 1 0 1 3 7	N / A	1 9 9 8 A D R (A u s t r a l i a) o n l y
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10395	1113	10193	1028	ISX-450	450	N/A	1650	1200	458	1700	450	1800	2814	1020	P102	N/A	N10259	N10103	N11607	Rev 2 FSC Conversion for ECM Code N10137 only
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L 8048 Calibration Matrix

S C	D O	F R	A P	M o d e l N a m e	A d v. (h p)	A d v. (r p m)	T · P. (l b - f t)	T · P. (r p m)	R a t e d (h p)	R a t e d (r p m)	G o v. (h p)	G o v. (r p m)	C P L (P P)	P W	F C	E C M C o d e f o r 3 6 8 0 5 0 9	E C M C o d e f o r 3 4 0 8 4 2 6	E C M C o d e f o r 3 6 8 1 4 0 5	E C M C o d e f o r 3 4 0 8 5 0 1	C o m m e n t s
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A blank coordinate grid with a horizontal axis labeled x and a vertical axis labeled y . The grid consists of 10 columns and 10 rows of squares. The x -axis is labeled from 1 to 10, and the y -axis is labeled from 1 to 10.

1 0 3 2 9	1 0 7 8	1 0 1 9 3	1 0 2 7	I S X - 4 5 0	4 5 0	N / A	1 6 5 0	1 2 0 0	4 5 8	1 7 0 0	4 5 0	1 8 0 0	2 4 4 0	1 0 0 8	P 0 2 2	N 1 0 0 2 1	N / A	N 1 0 1 3 7	N / A	1 9 9 8 A D R (A u s t r a l i a) o n l y
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10599	1113	10193	1028	ISX-450	450	N/A	1650	1200	458	1700	450	1800	8048	1021	P130	N/A	N10505	N10527	N11305	Rev 2 FSC Conversion for ECM Code N10137 only
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L 2 8 1 4 C a l i b r a t i o n M a t r i x

S C	D O	F R	A P	M o d e l N a m e	A d v. (h p)	A d v. (r p m)	T · P. (l b - f t)	T · P. (r p m)	R a t e d (h p)	R a t e d (r p m)	G o v. (h p)	G o v. (r p m)	C P L (P P)	P W	F C	E C M C o d e f o r 3 6 8 1 4 0 4	E C M C o d e f o r 3 4 0 8 4 2 6	E C M C o d e f o r 3 6 8 1 4 0 5	E C M C o d e f o r 3 4 0 8 5 0 1	C o m m e n t s
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1 0 3 5 4	1 0 9 7	1 0 1 9 3	1 0 2 8	I S X - 4 5 0	4 5 0	N / A	1 6 5 0	1 2 0 0	4 5 8	1 7 0 0	4 5 0	1 8 0 0	2 8 0 3	1 0 0 8	P 0 7 3	N 1 0 0 7 0	N / A	N 1 0 0 8 0	N / A	1 9 9 9 A D R (A u s t r a l i a) o n l y
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10395	1113	10193	1028	ISX-450	450	N/A	1650	1200	458	1700	450	1800	2814	1020	P102	N/A	N10259	N10103	N11607	Rev 2 FSC Conversion for ECM Code N10080 only
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L 8048 Calibration Matrix

S C	D O	F R	A P	M o d e l N a m e	A d v. (h p)	A d v. (r p m)	T . P. (l b - f t)	T . P. (r p m)	R a t e d (h p)	R a t e d (r p m)	G o v. (h p)	G o v. (r p m)	C P L (P P)	P W	F C	E C M C o d e f o r 3 6 8 1 4 0 4	E C M C o d e 3 4 0 8 4 2 6	E C M C o d e f o r 3 6 8 1 4 0 5	E C M C o d e f o r 3 4 0 8 5 0 1	C o m m e n t s
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A blank sheet of white graph paper with a light gray grid. The grid consists of small squares formed by thin gray lines. There are 20 columns and 15 rows of squares visible on the page.

1 0 3 5 4	1 0 9 7	1 0 1 9 3	1 0 2 8	I S X - 4 5 0	4 5 0	N / A	1 6 5 0	1 2 0 0	4 5 8	1 7 0 0	4 5 0	1 8 0 0	2 8 0 3	1 0 0 8	P 0 7 3	N 1 0 0 7 0		N 1 0 0 8 0		1 9 9 9 A D R (A u s t r a l i a) o n l y
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10599	1113	10193	1028	ISX-450	450	N/A	1650	1200	458	1700	450	1800	8048	1021	P130	N/A	N10505	N10527	N11305	Rev 2 FSC Conversion for ECM Code N10080 only
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10328	1094	10290	1027	Signature 500	500	N/A	1850	1200	525	1800	511	2000	2442	1007	P029	N10020	N10136	N/A	1998ADR (Australia) only (Power Curve 0)
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1 0 3 2 8	1 0 9 4	1 0 1 4 9	1 0 2 7	Si g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 4 4 2	1 0 0 7	P 2 5 2	N 1 0 0 2 0	N 1 0 1 3 6	N / A	1 9 9 8 A u s t r a l i a o n l y (P o w e r C u r v e 1)
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10328	1094	10151	1027	Signature 565	565	N/A	1850	1200	590	1800	565	2000	2442	1007	P014	N10020	N10136	N/A	1998ADR (Australia) only (Power Curve 2)
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10803	1160	10290	1027	Signature 500	500	N/A	1850	1200	525	1800	511	2000	8172	1077	P249	N11276	N11283	N11635	Rev 2 FSC conversion for ECM Codes N10020 and N10136
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1 0 2 3 5	1 0 7 7	1 0 1 5 1	1 0 2 7	Si g n a t u r e 5 6 5	5 6 5	N / A	1 8 5 0	1 2 0 0	5 9 0	1 8 0 0	5 6 5	2 0 0 0	2 4 4 2	1 0 0 2	P 0 1 4	N 1 0 0 1 5	N 1 0 1 3 5	N / A	N o n e
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10804	1077	10151	1027	Signature 565	N / A	1850	1200	590	1800	565	2000	8172	1002	P250	N11277	N11284	N11636	Rev 2 FSC conversion for ECM Codes N10015 and N10135
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1 0 5 5 6	1 1 6 0	1 0 1 5 1	1 0 2 7	Si g n a t u r e 5 6 5	5 6 5	N / A	1 8 5 0	1 2 0 0	5 9 0	1 8 0 0	5 6 5	2 0 0 0	2 4 4 2	1 0 5 2	P 1 1 4	N 1 0 2 2 3	N 1 0 2 2 5	N / A	1 9 9 8 / 9 9 A D R (A u s t r a l i a) o n l y
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10805	1160	10151	1027	Signature 565	N/A	1850	1200	590	1800	565	2000	8172	1052	P251	N11278	N11285	N11637	Rev 2 FSC conversion for ECM Codes N10223 and N10225
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1 0 2 3 4	1 0 7 7	1 0 1 4 9	1 0 2 7	Si g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 4 4 2	1 0 0 1	P 0 1 2	N 1 0 0 1 4	N 1 0 1 3 4	N / A	N o n e
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10806	1077	10149	1027	Signature 600	600	N/A	2050	1200	625	1800	600	2000	8172	1001	P252	N11279	N11286	N11638	Rev 2 FSC Conversion for ECM Codes N10014 and N10134
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1 0 2 3 4	1 1 8 5	1 0 1 4 9	1 0 2 7	Si g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 4 4 2	1 0 0 1	P 2 5 2	N 1 0 4 2 8	N / A	N / A	M e x i c o
1 0 8 0 6	1 1 8 5	1 0 1 4 9	1 0 2 7	Si g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	8 1 7 2	1 0 0 1	P 2 5 2	N 1 1 2 8 0	N 1 1 2 8 7	N 1 1 6 3 9	R e v 2 F S C o n v e r s i o n f o r E C M C o d e N 1 0 4 2 8

1 0 5 5 5	1 1 6 0	1 0 1 4 9	1 0 2 7	Si g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 4 4 2	1 0 5 1	P 1 1 3	N 1 0 2 2 2	N 1 0 2 2 4	N / A	1 9 9 8 / 9 9 A D R (A u s t r a l i a) o n l y
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10807	1160	10149	1027	Signature 600	600	N/A	2050	1200	625	1800	600	2000	8172	1051	P253	N11281	N11288	N11640	Rev 2 FSC Conversion for ECM Codes N10222 and N10224
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1 0 3 7 2	1 1 1 6	1 0 3 1 4	1 0 2 7	6 0 0 S e r v i c e	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	5 8 2	2 1 0 0	2 4 4 2	1 0 1 2	P 0 7 5	N 1 0 0 7 8	N 1 0 1 4 1	N / A	C a n a d a o n l y
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10808	1116	10314	1027	600 Service	600	N / A	2050	1200	625	1800	582	2100	8172	1012	P254	N11282	N11289	N11641	Rev 2 FSC conversion for ECM Codes N10078 and N10141
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C P L 2 8 3 4 / 2 8 3 6																				
S C	D O	F R	A P	M o d e l N a m e	A d v. (h p)	A d v. (r p m)	T · P. (l b - f t)	T · P. (r p m)	R a t e d (h p)	R a t e d (r p m)	G o v. (h p)	G o v. (r p m)	C P L (P P)	P W	F C	E C M C o d e f o r 3 6 8 1 4 0 4	E C M C o d e f o r 3 6 8 1 4 0 5	E C M C o d e f o r 3 4 0 8 4 2 6	E C M C o d e f o r 3 4 0 8 5 0 1	C o m m e n t s
1 0 3 7 4	1 1 0 4	1 0 2 9 7	1 0 3 4	I S X 5 6 5	5 6 5	N / A	1 8 5 0	1 2 0 0	5 9 0	1 8 0 0	5 6 5	2 0 0 0	2 6 3 0	1 0 1 4	P 0 5 9	N 1 0 0 7 4	N 1 0 1 2 0	N / A	N / A	

10426	1104	10297	1034	ISX565	565	N/A	1850	1200	590	1800	565	2000	2834	1014	P086	N/A	N10184	N10274	N11250	Rev2FSConve rsionforECMC odesN10074 andN10120
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1 0 3 7 3	1 1 0 4	1 0 3 1 1	1 0 3 4	I S X 6 0 0	6 0 0	N / A	1 8 5 0	1 2 0 0	6 1 0	1 8 0 0	6 0 0	2 0 0 0	2 6 3 0	1 0 1 3	P 0 7 6	N 1 0 0 7 3	N 1 0 1 1 9	N / A	N / A	
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10425	1104	10311	1034	ISX600	600	N/A	1850	1200	610	1800	600	2000	2834	1013	P085	N/A	N10168	N10272	N11247	Rev2FSConversions for ECM Codes N10073 and N10119
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10409	1096	10290	1028	S i g n a t u r e 5 0 0	500	N / A	1850	1200	525	1800	511	2000	2801	1031	P072	N10133	N10114	N / A	N / A	A D R (A u s t r a l i a) o n l y
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10435	1096	10290	1028	S i g n a t u r e 5 0 0	500	N / A	1850	1200	525	1800	511	2000	2836	1031	P095	N / A	N10180	N10282	N11055	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 1 3 3 a n d N 1 0 1 1 4
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1 0 3 3 3	1 0 9 6	1 0 1 5 1	1 0 3 4	S i g n a t u r e 5 6 5	5 6 5	N / A	1 8 5 0	1 2 0 0	5 9 0	1 8 0 0	5 6 5	2 0 0 0	2 6 3 0	1 0 0 2	P 0 3 1	N 1 0 0 6 0	N 1 0 1 1 6	N / A	N / A	
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10422	1096	10151	1034	S i g n a t u r e 5 6 5	565	N / A	1850	1200	590	1800	565	2000	2834	1002	P082	N / A	N10175	N10269	N11615	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 0 6 0 a n d N 1 0 1 1 6
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10408	1096	10151	1028	S i g n a t u r e 5 6 5	565	N / A	1850	1200	590	1800	565	2000	2801	1030	P071	N10132	N10113	N / A	N / A	A D R (A u s t r a l i a) o n l y
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10434	1096	10151	1028	S i g n a t u r e 5 6 5	565	N / A	1850	1200	590	1800	565	2000	2836	1030	P094	N / A	N10179	N10281	N11295	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 1 3 2 a n d N 1 0 1 1 3
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1 0 3 9 6	1 1 0 4	1 0 3 1 5	1 0 3 4	5 6 5 S T X	5 6 5	N / A	1 8 5 0	1 2 0 0	5 9 0	1 8 0 0	5 6 5	2 0 0 0	2 6 3 0	1 0 2 7	P 0 7 8	N 1 0 0 7 7	N 1 0 1 2 7	N / A	N / A	
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10424	1104	10315	1034	565STX	565	N/A	1850	1200	590	1800	565	2000	2834	1027	P084	N/A	N10167	N/A	N/A	Rev2FSC Conversion for ECM Codes N10077 and N10127
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1 0 3 3 2	1 0 9 6	1 0 1 4 9	1 0 3 4	S i g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 6 3 0	1 0 0 1	P 0 3 0	N 1 0 0 4 0	N 1 0 1 1 5	N / A	N / A	
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10421	1096	10149	1034	S i g n a t u r e 6 0 0	600	N / A	2050	1200	625	1800	600	2000	2834	1001	P081	N / A	N10174	N10268	N11243	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 0 4 0 a n d N 1 0 1 1 5
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1 0 3 3 2	1 1 8 6	1 0 1 4 9	1 0 3 4	S i g n a t u r e 6 0 0	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 6 3 0	1 0 0 1	P 0 3 0	N 1 0 4 8 8	N 1 0 4 3 0	N / A	N / A	M e x i c o n l y
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10421	1186	10149	1034	S i g n a t u r e 6 0 0	600	N / A	2050	1200	625	1800	600	2000	2834	1001	P081	N / A	N10437	N10426	N11614	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 4 8 8 a n d N 1 0 4 3 0
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10407	1096	10149	1028	S i g n a t u r e 6 0 0	600	N / A	2050	1200	625	1800	600	2000	2801	1029	P070	N10131	N10112	N / A	N / A	A D R (A u s t r a l i a) o n l y
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10433	1096	10149	1028	S i g n a t u r e 6 0 0	600	N / A	2050	1200	625	1800	600	2000	2836	1029	P093	N / A	N10186	N10280	N11260	R e v 2 F S C o n v e r s i o n f o r E C M C o d e s N 1 0 1 3 1 a n d N 1 0 1 1 2
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1 0 3 5 1	1 1 0 4	1 0 2 9 6	1 0 3 4	6 0 0 S T X	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	6 0 0	2 0 0 0	2 6 3 0	1 0 1 1	P 0 5 8	N 1 0 0 6 2	N 1 0 1 2 3	N / A	N / A	
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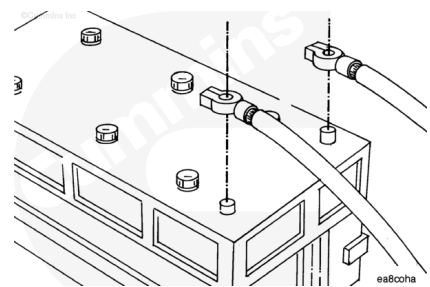
10423	1104	10296	1034	600STX	600	N/A	2050	1200	625	1800	600	2000	2834	1011	P083	N/A	N10166	N10270	N11245	Rev2FSConve rsionforECMC odesN10062 andN10123
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1 0 3 7 1	1 1 1 5	1 0 3 1 4	1 0 3 4	6 0 0 S e r v i c e	6 0 0	N / A	2 0 5 0	1 2 0 0	6 2 5	1 8 0 0	5 8 2	2 1 0 0	2 6 3 0	1 0 1 2	P 0 7 4	N 1 0 0 7 9	N 1 0 1 4 2	N / A	N / A	C a n a d a o n l y
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10440	1115	10314	1034	600 Service	600	N / A	2050	1200	625	1800	582	2100	2834	1012	P100	N / A	N10188	N10321	N11618	Rev 2 FSC Conversion for ECM Codes N10079 and N10142
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 WARNING 

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



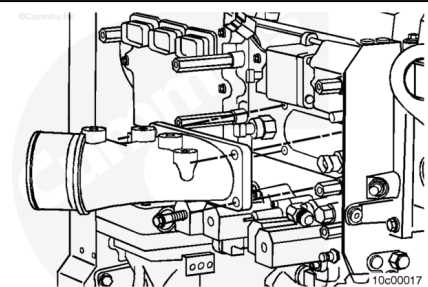
Disconnect the batteries. Refer to Procedure Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/10/10-013-009.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

Remove the following:

- Air inlet piping
- Fuel control covers
- Four-step wastegate controller wires (if equipped), hose, mounting capscrews, and controller (if equipped).

The fuel shutoff wire can also be disconnected at this time, to make installation of the wiring kit easier.

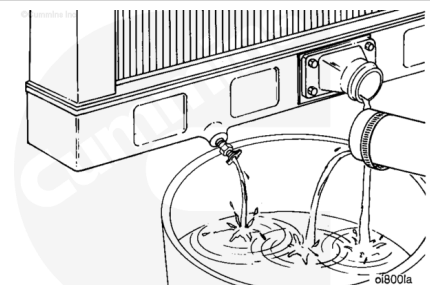
Remove the air intake connection. Refer to Procedure Refer to Procedure 010-080 (/qs3/pubsys2/xml/en/procedures/10/10-010-080.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.



⚠ WARNING ⚠

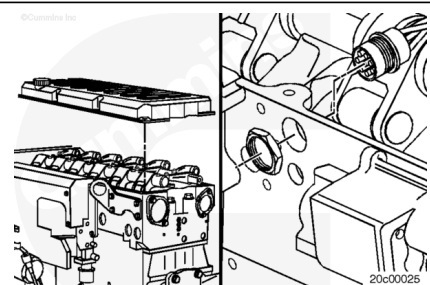
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the engine coolant. Refer to Procedure Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.



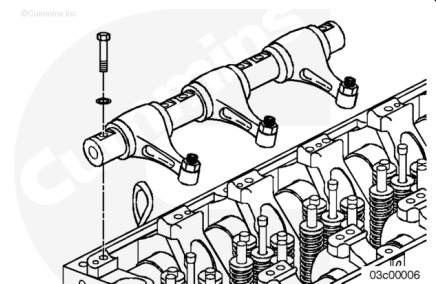
Remove the rocker lever cover. Refer to Procedure Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/10/10-003-011.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

Disconnect the engine brake solenoid wiring harness. Refer to Procedure Refer to Procedure 020-015 (/qs3/pubsys2/xml/en/procedures/10/10-020-015.html) in the



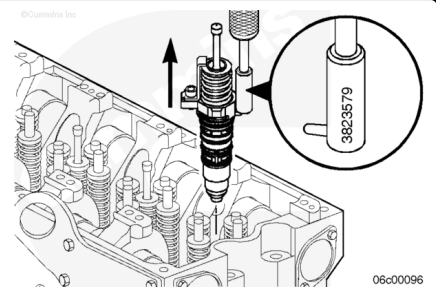
Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

Remove the rocker lever shaft assemblies. Refer to Procedure Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/10/10-003-009.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.



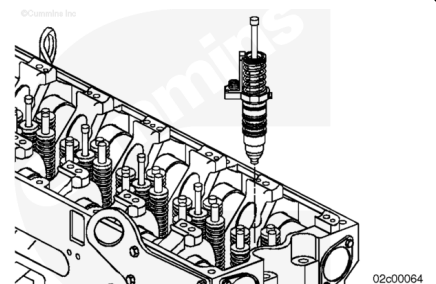
Remove the injectors. Refer to Procedure Refer to Procedure 006-026 (/qs3/pubsys2/xml/en/procedures/10/10-006-026.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

See Table 1 for the correct injector part number for your application.

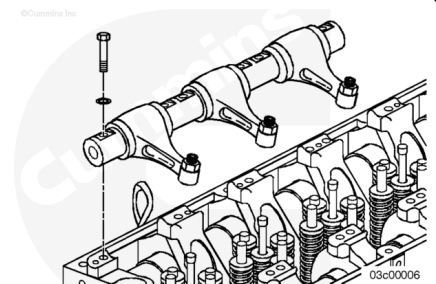


Install the injectors. Refer to Procedure Refer to Procedure 006-026 (/qs3/pubsys2/xml/en/procedures/10/10-006-026.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

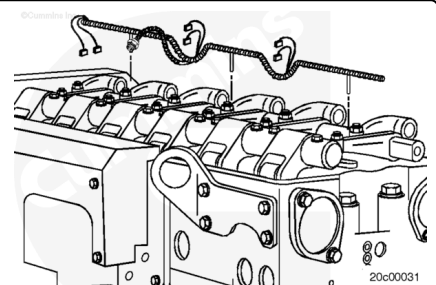
See Table 1 for the correct injector part number for your application.



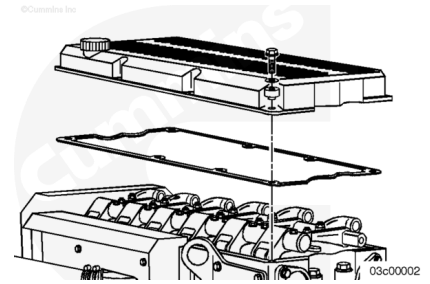
Install the rocker lever shaft assemblies. Refer to Procedure Refer to Procedure 003-009 (/qs3/pubsys2/xml/en/procedures/10/10-003-009.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.



Connect the engine brake solenoid wiring harness. Refer to Procedure Refer to Procedure 020-015 (/qs3/pubsys2/xml/en/procedures/10/10-020-015.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

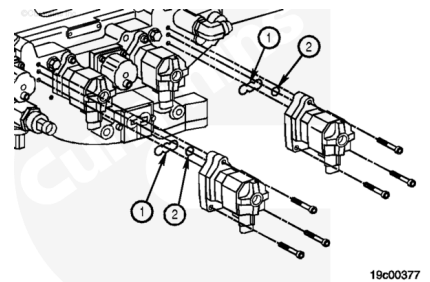


Install the rocker lever cover. Refer to Procedure Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/10/10-003-011.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

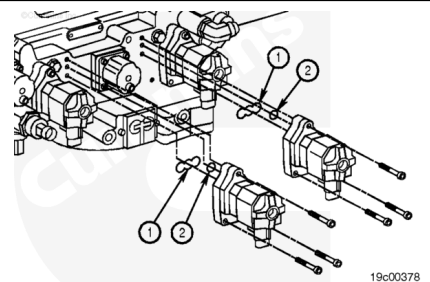


Clean the area around the fueling and timing actuators.

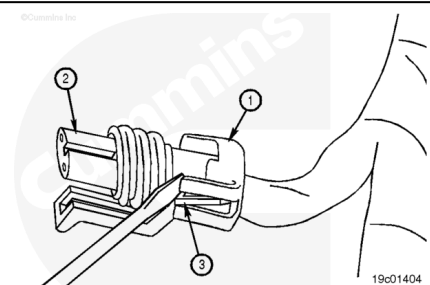
Remove the fueling actuators. See Procedure 019-103 in the appropriate electronic controls troubleshooting and repair manual.



Remove the timing actuators. See Procedure 019-104 in the appropriate electronic controls troubleshooting and repair manual.



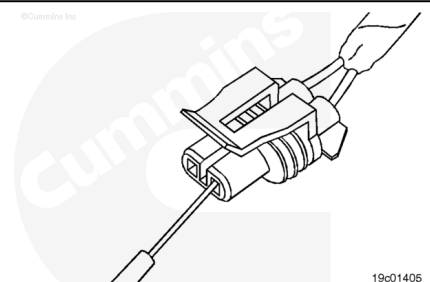
Remove the Metri-Pack connector wire lock (1) from the wiring harness connector (2) by pulling the locking tabs (3) apart on the wire lock.



Hold the connector with the lock tab up, insert the Metri-Pack extraction tool, Part Number 3164859, on the top side of the terminal until it stops.

Grasp the wire at the rear of the connector and remove it from the connector body.

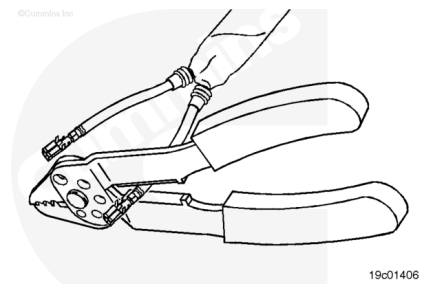
Repeat this step for each actuator connector.



Grasp the terminal seal and slide it back on each wire on each actuator connector.

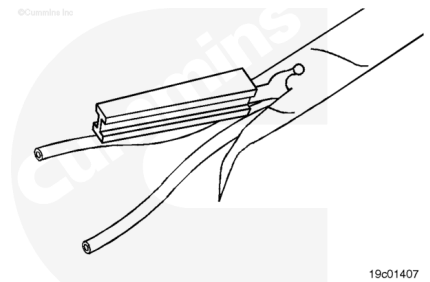
Use wire crimp tool, Part Number 3822930, to cut the wire as close to the terminal body as possible.

Remove and discard the wire seals.

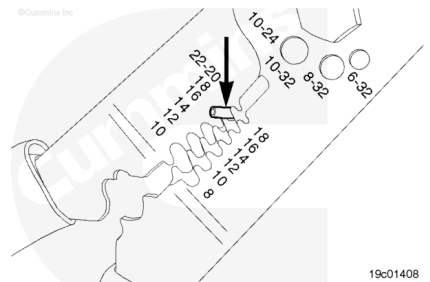


Use service tool, Part Number 3822932, remove approximately 51 mm [2 in] of the wiring harness protective cover from the actuator leads to the main harness.

Repeat this step for each actuator lead set.



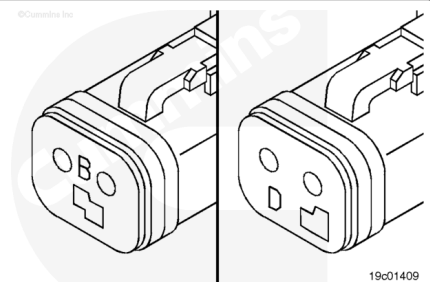
Use wire crimp tool, Part Number 3822930, to remove 6 mm [1/4 in] of insulation from each of the eight actuator wires.



Install the correct repair connector on the correct actuator wires, matching black to black and white to white.

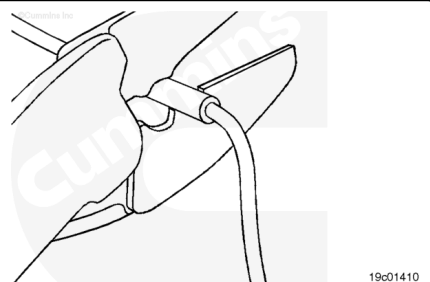
The bare wire **must** extend into the insulated butt splice connector.

Use the repair connector, Part Number 3164856, with the key "B" (black) on the two outer actuator connections and the repair connector, Part Number 3164857, with the key D (brown) on the two inner actuator connections.



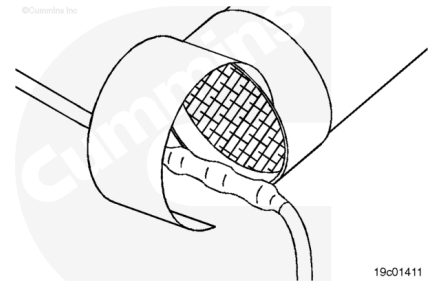
Use wire crimp tool, Part Number 3822930, to crimp the repair connectors onto each bare wire.

Repeat this step for each actuator repair connector.



Use a heat gun, Part Number 3822860, or an open flame to heat the shrink tubing. The tubing will shrink and make the connection waterproof.

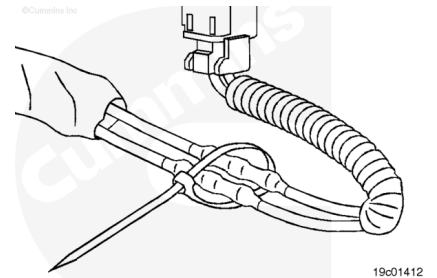
Repeat this step for each actuator repair connector.



Place a length of convolute tubing, Part Number 3164858, over each set of wires that were repaired.

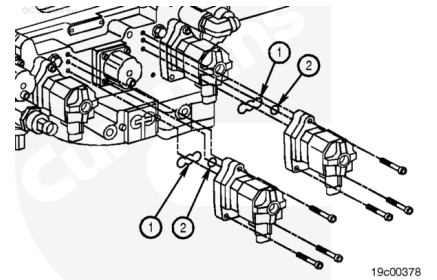
Use small wire ties to secure the two butt splice connectors from each repair to the main body of the engine harness.

At least 102 mm [4 in] of the repair connector **must** remain free to plug into the actuators.



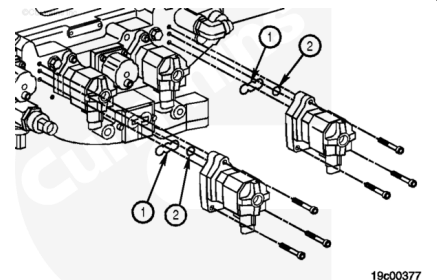
Install the new timing actuators. See Procedure 019-104 in the appropriate electronic controls troubleshooting and repair manual.

See Table 1 for the correct actuator part numbers.



Install the new fueling actuators. See Procedure 019-103 in the appropriate electronic controls troubleshooting and repair manual.

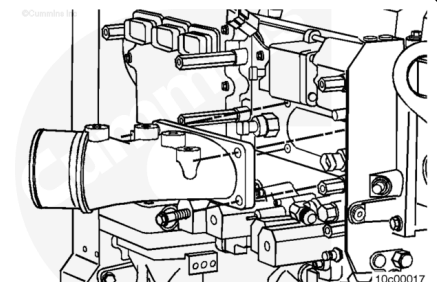
See Table 1 for the correct actuator part numbers.



Install the air intake connection. Refer to Procedure 010-080 (</qs3/pubsys2/xml/en/procedures/10/10-010-080.html>) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

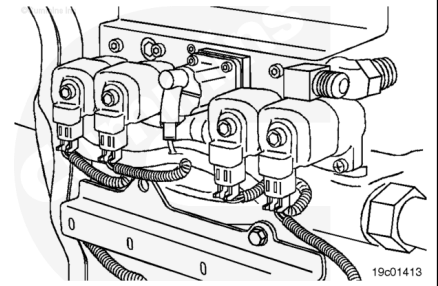
Install the four-step wastegate controller wires (if equipped), hose, mounting capscrews, and controller (if equipped).

Connect the fuel shutoff control wire.

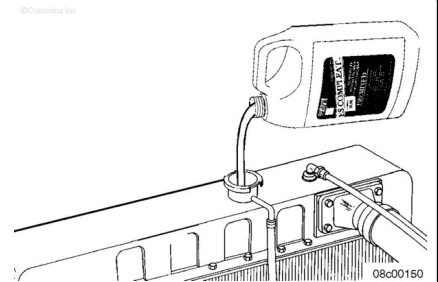


Verify the installation of components and that all wires are secure and free from abrasion.

Verify the location of installed components.

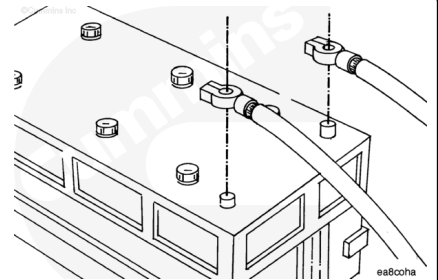


Fill the cooling system. Refer to Procedure Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.



⚠ WARNING ⚠

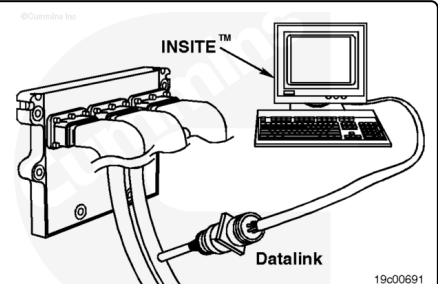
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Connect the batteries. Refer to Procedure Refer to Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/10/10-013-009.html) in the Troubleshooting and Repair Manual, Signature, ISX and QSX15 Engines, Bulletin 3666239.

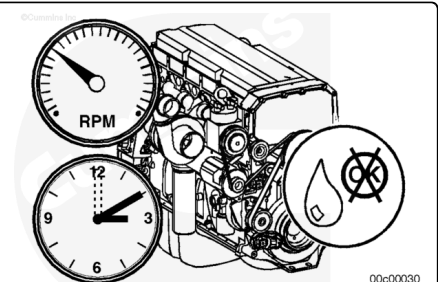
The ECM can now be calibrated with the correct calibration for the conversion from Rev 1 to Rev 2 fuel system.

See Table 2 for the correct calibration for your application.



Operate the engine for 8 to 10 minutes to check for correct engine operation, unusual noises, and coolant, fuel, or lubricating oil leaks.

Repair all leaks and component problems. See the appropriate procedures in the appropriate manual.



Document History

Date	Details
2017-03-27	Module Converted
2017-4-10	Update to format

Last Modified: 18-May-2017
